



PREDICTING TENNESSEE CROP YIELDS

A Machine Learning Approach using Weather, Soil and
Historical Data

CORN ■ SOYBEAN ■ WHEAT

Tennessee county level Data
2000-2024



Project Overview:

Can Weather, Soil, crop's history predict how much corn, soybean, and wheat will yield next season, across Tennessee?

Problem:

Crop yield is affected by numerous environmental factors, making accurate prediction challenging.

Goal:

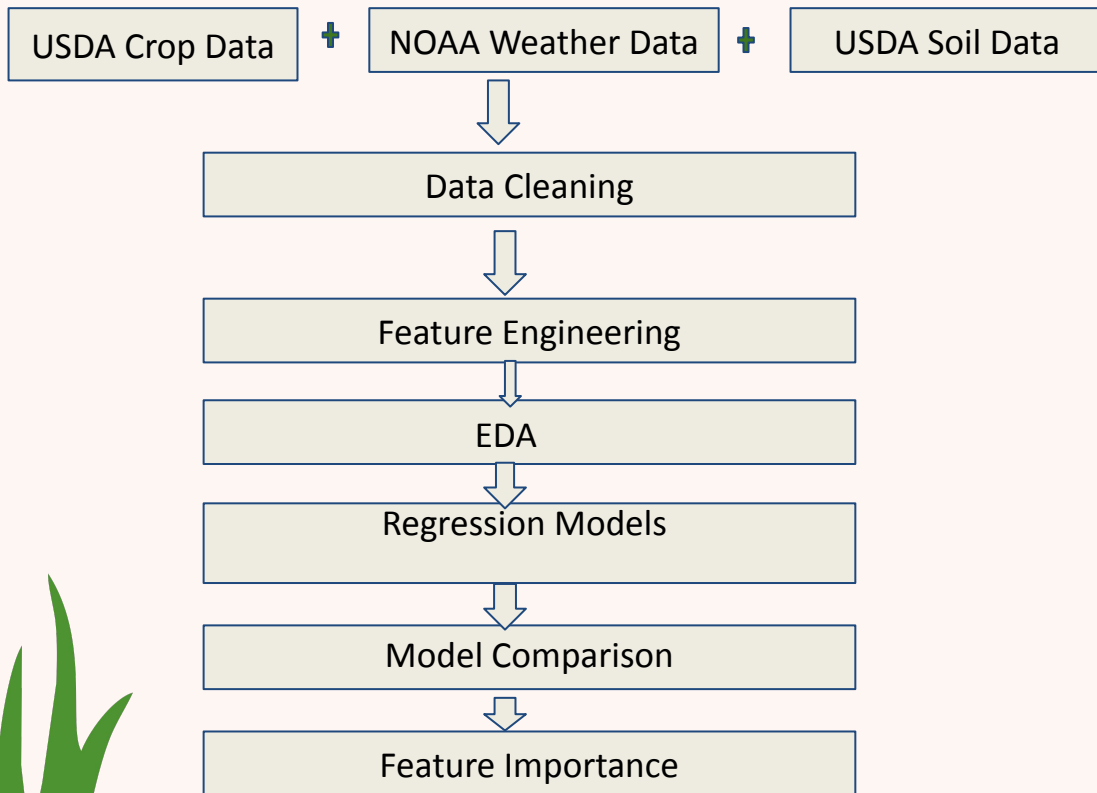
Develop machine learning regression models to predict crop yield using

- Historical crop yield
- Weather data
- Soil characteristics

for Tennessee counties.



Project Workflow:



Data Gathering: Three Sources, One Dataset

Crop Yield

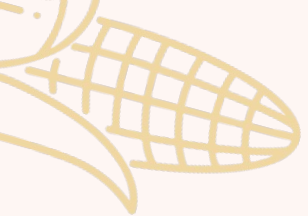
USDA NASS(API call)
Yield_bu_acre by
county, year, crop

Weather

NOAA (API call)
Precipitation(PRCP), MAX
Temp, Min Temp

Soil

USDA SSURGO(Geographical
Database contains soil type
and distribution)
Drainage, Soil ph, water
storage, texture

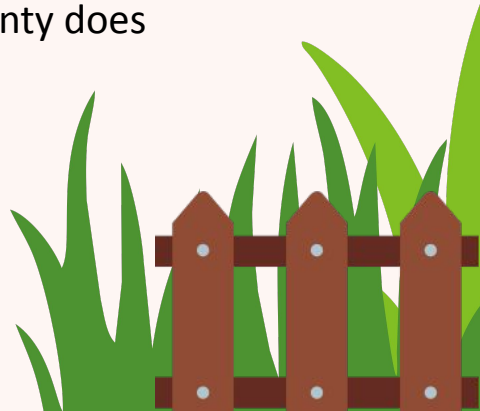


Data Integration:

Performed

- Spatial Joins
- County matching
- Weather station mapping

Used GeoPandas to assign weather stations to Tennessee counties and nearest-station matching where needed. (Only one County does not have the weather station , for that used nearest station)





Cleaning the Data:

Crop Data:

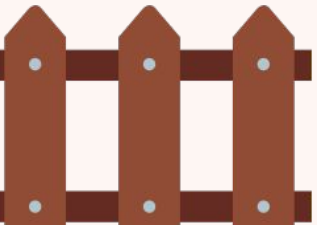
- Removed missing countie(1.1%) (total-3561 Rows)
- Converted yield to numeric
- Averaged duplicate county-year-crop records
- Standardized county names

Weather Data:

- Filled missing PRCP
- Filled TMAX
- Filled TMIN using median values
- Standardized county names

Soil Data:

- Renamed cryptic variables into descriptive feature names
- Improved readability for modeling





Feature Engineering:

Created meaningful features to help improve model performance

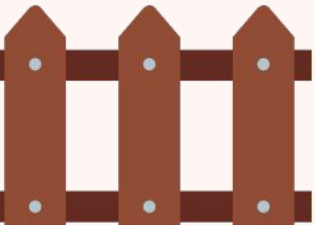
Previous Year's Yield (`yield_bu_acre_lag1`)

- Previous year's yield for the same county and crop.
- Captures historical production trends.
- Provides context about long-term farming conditions.

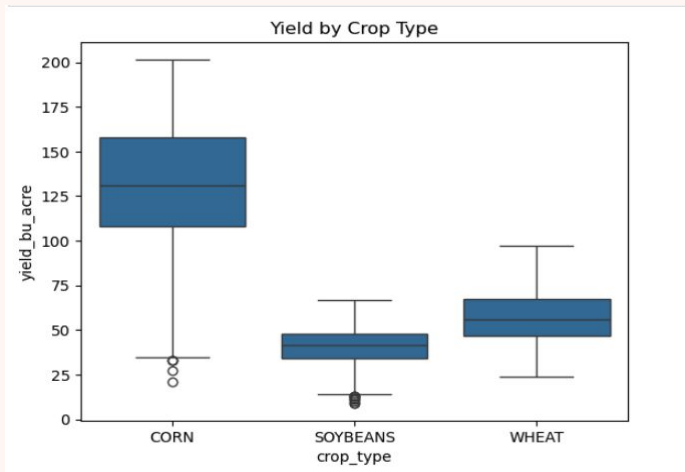
Monthly Weather Features

- Monthly precipitation (PRCP)
- Monthly maximum temperature (TMAX)
- Monthly minimum temperature (TMIN)
- Growing seasons were customized by crop

Crop	Critical Month	Growing Season
Corn	July	June-August
Soybeans	August	July-September
Wheat	May	April-June



Distribution of Crop Yield by Crop Type:



- Yield distributions differ across crop types.
- Variability exists within each crop.
- Crop type is an important predictor and was included as a model feature.

Note: Yield is measured in **bushels per acre**, and expected yield ranges differ among crop types. The plot illustrate variability within each crop rather than compare productivity across crops.

Ridge Regression Model

My first model was a Ridge Regression model, which served as a baseline.

Data Preparation

Training Data: 2000-2017

Unseen Data: 2018-2024

Features: Yearly Weather, Soil Characteristics , Historical Yield, county, crop

Evaluation metrics: R2, mse, mae

Pipeline

- StandardScaler
- OneHotEncoder
- TimeSeriesSplit Cross Validation

Baseline model showed limited predictive performance

	ridge_unseen_r2	ridge_unseen_mse	ridge_unseen_mae	cv_r2_mean	\
CORN	-0.846869	1131.001878	29.149757	-1.054111	
SOYBEANS	-0.808364	73.646149	7.223161	-1.530757	
WHEAT	-0.416367	139.616470	9.539095	-0.769605	
	naive_baseline_r2	naive_baseline_mae			
CORN	-0.570440	21.751685			
SOYBEANS	-0.464904	5.828963			
WHEAT	0.074935	7.654861			

Random Forest – Yearly Weather

Features

- Annual precipitation
- Annual temperature
- Soil variables
- County
- Crop

Random Forest can capture nonlinear relationships better than Ridge Regression.

Because weather and crop yield do not always have a simple linear relationship, Random Forest was expected to perform better

	rf_unseen_r2	rf_unseen_mse	rf_unseen_mae	internal_r2_test
CORN	-0.450186	888.077853	23.420745	-0.439895
SOYBEANS	-0.528818	62.261554	6.324024	-2.875580
WHEAT	-0.349926	133.067149	9.130951	-1.606776

	naive_baseline_r2
CORN	-0.570440
SOYBEANS	-0.464904
WHEAT	0.074935

Random Forest – Monthly Weather

Instead of using one annual precipitation value, included precipitation and temperature for each month.

The model captures weather during critical crop growth stages, allowing it to learn seasonal effects that yearly averages can miss.

	rf_unseen_r2	rf_unseen_mse	rf_unseen_mae	naive_baseline_r2	\
CORN	-0.006539	624.302166	20.980569	-0.571931	
SOYBEANS	0.027419	40.448575	4.891285	-0.445399	
WHEAT	-0.124758	112.129747	8.324217	0.077783	
	naive_baseline_mae	internal_r2_test			
CORN	21.857817	-0.231546			
SOYBEANS	5.826537	-2.563103			
WHEAT	7.690714	-2.428720			

Feature Importance

WHEAT

feature	importance
num__yield_bu_acre_lag1	0.396417
num__PRCP	0.118499
num__temp_range	0.087900
num__TMAX	0.039565
num__avg_temp	0.029175
num__drain_CD	0.028436
num__TMIN	0.027518
num__drain_D_very_low	0.027143
num__water_storage_0_100in	0.018781
num__water_storage_0_150in	0.017128
num__soil_ph_avg	0.016276
num__drain_A_high	0.014985
num__drain_BD	0.013012
num__drain_C_low	0.010903
num__water_table_depth_avg	0.010367

SOYBEANS

feature	importance
num__yield_bu_acre_lag1	0.264104
num__PRCP	0.258236
num__TMAX	0.084665
num__temp_range	0.048349
num__avg_temp	0.028914
num__TMIN	0.028803
num__organic_matter_avg	0.023320
num__wettest_well	0.017927
num__drain_B_mod	0.016475
num__drainage_well	0.014902
num__drain_C_low	0.014199
num__drain_A_high	0.013432
num__soil_ph_avg	0.012078
num__drain_D_very_low	0.011703
num__bedrock_depth_min	0.011191

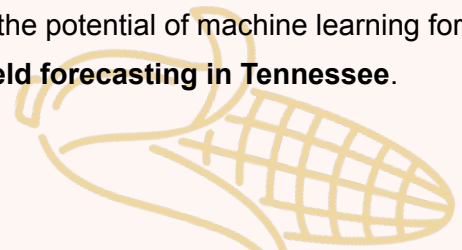
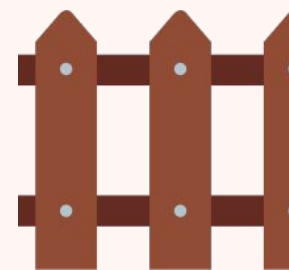
CORN

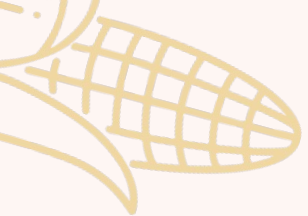
feature	importance
num__yield_bu_acre_lag1	0.304230
num__PRCP	0.219453
num__TMAX	0.084940
num__temp_range	0.056600
num__avg_temp	0.030748
num__TMIN	0.027260
num__drain_CD	0.026706
num__drain_D_very_low	0.018407
num__bedrock_depth_min	0.017080
num__water_storage_0_150in	0.016688
num__soil_ph_avg	0.015705
num__drain_B_mod	0.013428
num__organic_matter_avg	0.012440
num__drain_BD	0.011105
num__drain_C_low	0.011002

Weather variables, particularly precipitation and monthly temperature, together with soil characteristics, and historical crop yield produces better predictions than using only one type of data.



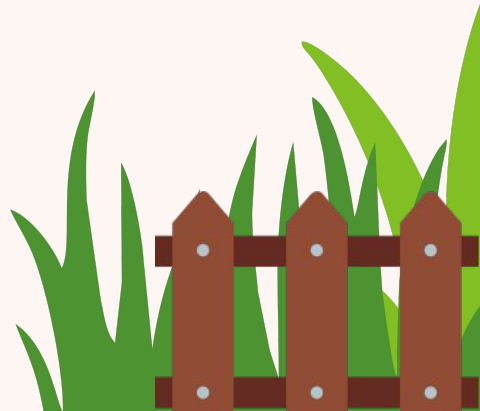
Conclusion

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- Successfully integrated **crop yield, weather, and soil datasets** into a unified predictive framework.
 - Adding **monthly weather features improved soybean prediction**, showing that weather timing during crop growth stages provides useful information. However, improvements were limited for corn and wheat, indicating that **weather alone cannot fully explain yield variation**.
 - Feature importance analysis showed that **previous year's yield, weather conditions, and soil characteristics** were the most influential predictors.
 - For **wheat, previous year's yield was the dominant factor**, suggesting that historical production trends and crop management practices may have a stronger impact than annual weather variations.
 - The framework demonstrates the potential of machine learning for **data-driven agricultural planning and yield forecasting in Tennessee**.



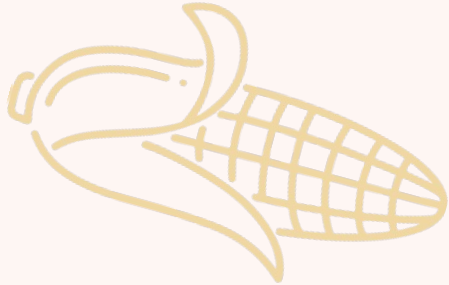
Future Work

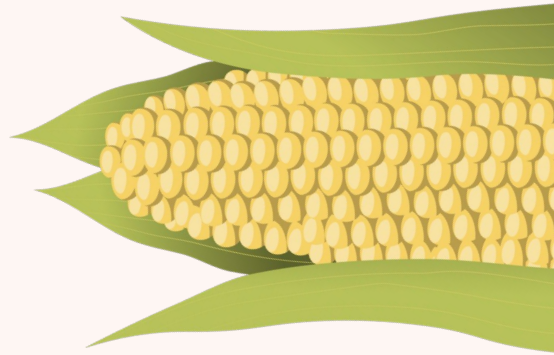


- Evaluate additional algorithms (e.g., Gradient Boosting or XGBoost)
 - Develop an interactive dashboard for crop yield prediction and visualization
 - Expand the analysis to additional states and crop types
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Questions/Suggestions





Thank You



Model Comparison:

Model	Strengths	Weaknesses
Ridge Regression	Lower Mse for corn and wheat among yearly models. Simple and interpretable	Mostly negative R^2
RF – Yearly	Captures nonlinear effects	Highest prediction errors; negative R^2
RF – Monthly	Best overall performance; positive R^2 . lowest errors	Still modest predictive power